

l_1 -embeddability of Regular Quadrilateral Maps on the Torus and the Klein bottle *

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Abstract

A connected graph G is said to be l_1 -embeddable if its graph metric admits an isometric embedding into an l_1 -space. The graphs $Q_{k,m,r}$ and $Q_{k,m,e}$ form two families of regular quadrilateral maps on the torus, obtained from suitable identifications of the Cartesian product $P_m \square P_k$ ($k, m \geq 1$). The graphs $Q_{k,m,a}$, $Q_{k,m,b}$, and $Q_{k,m,h}$ form three families of 4-regular quadrilateral maps on the Klein bottle, obtained from suitable identifications of $P_m \square C_k$ ($k \geq 1, m \geq 2$). In this paper, we completely characterize the l_1 -embeddable graphs in these five families. More precisely, all graphs $Q_{k,m,r}$, $Q_{k,m,e}$, $Q_{k,m,a}$, $Q_{k,m,b}$, and $Q_{k,m,h}$ are not l_1 -embeddable, with the only exceptions being $Q_{k,m,0}$, $Q_{k,2,1}$, $Q_{k,1,e}$, $Q_{5,2,e}$, $Q_{2i,2,e}$ ($i \in \mathbb{Z}_{>0}$), $Q_{2,2,a}$, $Q_{4,2,a}$, $Q_{2,m,b}$, and $Q_{1,2,h}$.

Key words: l_1 -embeddable, regular quadrilateral maps, torus, Klein bottle

1 Introduction

The graphs discussed in this article are all finite, connected, and undirected. For a graph G , its vertex set is denoted by $V(G)$ and its edge set is denoted by $E(G)$. A *path* in a graph G is defined as a finite non-empty sequence $w = v_0 e_1 v_1 e_2 v_2 \dots e_n v_n$, where the elements of the sequence alternate between vertices and edges, and all vertices and edges in the sequence are distinct, such that for $1 \leq i \leq n$, the endpoints of the edge e_i are v_{i-1} and v_i . The sequence P is called a path from v_0 to v_n , or a v_0, v_n -path. The vertices v_0 and v_n are referred to as the starting and ending vertices of P , respectively, while the vertices $v_1, v_2, \dots, v_{n-1}$ are considered as its internal vertices. The number of edges contained in P , is called the length of P . The *distance* between two vertices u and v in a connected graph G is the minimum length among all u, v -paths, denoted by $d_G(u, v)$. When G is clear from the context, simply write as $d(u, v)$. If $d_H(x, y) = d_G(x, y)$ for any two vertices of H , we say that H is an isometric subgraph of G . A cycle is a path whose origin and terminus are identical.

Based on the concept of distance, we begin by understanding what a *metric space* is. Given a set of real numbers X and the set of non-negative real numbers $\mathbb{R}^+$. A function

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$d : X \times X \rightarrow \mathbb{R}^+$ is defined to be a distance on X if it satisfies symmetry and for all $i \in X$ $d(i, i) = 0$. If d also satisfies the triangle inequality, then d is called a *semi-metric* on X and (X, d) is called a metric space. Furthermore, if $d(i, j) = 0$ holds only for $i = j$, then d is called a *metric* on X .

The vector space $\mathbb{R}^n$ associated with the metric for any $x, y \in \mathbb{R}^n$, $d_{l_1}(x, y) = \sum_{i=1}^n |x_i - y_i|$ is called the l_1 -space. Let $G = (V, E)$ be a connected graph. Clearly, d_G is a metric on $V(G)$, and the metric space $(V(G), d_G)$ is called the *graph metric space* associated with G . A graph G is said to be l_1 -embeddable (or an l_1 -graph) if its associated metric space can be isometrically embedded into the l_1 -space. In other words, there exists a mapping $\phi : V(G) \rightarrow \mathbb{R}^n$ such that for all vertices x, y of $V(G)$,

$$d_G(x, y) = d_{l_1}(\phi(x), \phi(y)).$$

According to the monograph [5], cycles, complete graphs, trees, hypercubes, etc., are l_1 -graphs.

Throughout this paper, we use the notation $G \cong H$ to indicate that the graph G is isomorphic to the graph H .

The *Cartesian product* $H_1 \square H_2$ of graphs H_1 and H_2 is the graph with vertex set $H_1 \times H_2$ such that the vertex (u_1, u_2) is adjacent to the vertex (v_1, v_2) whenever either $u_1 = v_1$ and u_2 is adjacent to v_2 in H_2 , or u_1 is adjacent to v_1 in H_1 and $u_2 = v_2$.

The *n -dimensional hypercube graph* Q_n is defined as follows: let $\Omega = \{1, 2, \dots, n\}$. The vertex set of Q_n consists of all subsets of Ω . Two vertices A and B are adjacent if and only if their symmetric difference satisfies $|A \triangle B| = 1$, where $\triangle$ denotes the symmetric difference of sets (i.e., $A \triangle B = (A \setminus B) \cup (B \setminus A)$). It is known that $d_{Q_n}(A, B) = |A \triangle B|$ for any two vertices of Q_n .

According to [1], a graph G is an l_1 -graph if and only if there exist positive integers λ and k such that G can be λ -embedded into a hypercube Q_n , i.e., there is a mapping $\phi : V(G) \rightarrow V(Q_n)$ such that

$$\lambda \cdot d_G(u, v) = d_{Q_n}(\phi(u), \phi(v)),$$

for any two vertices u, v of G . Especially, if $\lambda=1$, then G is an *isometric subgraph* of Q_n (also name partial cube). Xie, Feng and Xu [19] characterizes 2-arc-transitive partial cubes. In addition, the characterizations of l_1 -graphs were given by Shpectorov [13], Grishukin and Deza [4]: a graph is an l_1 -graph if and only if it is an isometric subgraph of a Cartesian product of complete graphs, cocktail part graphs, and half-cubes. Shpectorov [13] showed that l_1 -graphs can be recognized in polynomial time. In [6], Deza and Shpectorov filled in the details and analyzed the complexity of the algorithm. In [18], Wang and Li studied the l_1 -embeddability problem for the graph obtained by gluing two non-bipartite l_1 -graphs along an edge.

Prisacaru, Soltan, and Chepoi [12] have proved that all planar graphs with a minimum face size of 5 and a minimum vertex degree of 4 are l_1 -embeddable. Chepoi, Deza, and Grishukhin [2] provided a criterion for determining whether a planar graph is an l_1 -graph. Apart from the plane, the other well-known surfaces include the sphere, cylinder, Möbius map, torus, and the Klein bottle. Wang and Zhang [16] proved that out of all the hexagonal and quadrilateral tiling graphs on the regular Möbius map, only $H_{2,2}$, $H_{3,3}$ and $Q_{2,2}$ are l_1 -graphs. Wang and Shpectorov [17] proved that for any l_1 -embeddable quadrilateral Möbius map, there exists a unique shortest nulhomotopic cycle. Zhang and Wang [21] proved that of all single-walled open-ended nanotubes, only those of type (1,0), (0,1) and (1,1) can be isometrically embedded into

hypercubes. Wang, Xiong and Chen [15] proved that among the shifted quadrilateral tiling graphs on the cylinder, only $O_{m,n,0}$ and $O_{m,3,1}$ are l_1 -embeddable.

This paper considers tilings that are locally akin to some of the plane tilings explored by Grünbaum and Shephard [8]. Thomassen [11] fully classified all regular tilings on the torus and the Klein bottle, identifying them as the 3-regular tiling by hexagons, the 4-regular tiling by quadrilaterals, and the 6-regular tiling by triangles. Deza and Shpectorov [7] classified all finite closed polyhexes (i.e., trivalent surface graphs with hexagonal faces) that are l_1 -graphs. But, the l_1 -embeddability of 4-regular quadrilateral tilings on the torus and the Klein bottle has not yet been studied. Equivalently, such tilings can be regraded as regular maps on these surfaces, where the notion of maps is the same as that defined in [17]. That means 4-regular quadrilateral tilings on the torus and the Klein bottle are 2-cell embeddings in the torus and the Klein bottle such that every face is a quadrilateral.

Thomassen [11] proved that there are two classes of 4-regular quadrilateral maps $Q_{k,m,r}$, $Q_{k,m,e}$ of the torus and six classes quadrilateral maps $Q_{k,m,a}$, $Q_{k,m,b}$, $Q_{k,m,c}$, $Q_{k,m,f}$, $Q_{k,m,g}$, $Q_{k,m,h}$ on the Klein bottle. In fact, as shown in [20], $Q_{k,m,a} \cong Q_{2m, \frac{k}{2}, a}$, $Q_{k,m,f} \cong Q_{2(m+1), \frac{k}{2}, b}$, $Q_{k,m,g} \cong Q_{2m+1, \frac{k}{2}, a}$. In this paper, we focus on recognizing which of these maps are l_1 -embeddable.

The structure of this paper is as follows: In Section 2, we present the label of l_1 -graphs and the necessary conditions. In Section 3, we studied the l_1 -embeddability of the two classes of regular quadrilateral maps on the torus, and deduced the l_1 -embeddability of a class of special circulant graphs. In Section 4, we describe the construction of the three classes of regular quadrilateral maps on the Klein bottle and determine which of them are l_1 -embeddable. In Section 5, we point out that the number of the l_1 -embeddable regular quadrilateral map on the torus and the Klein bottle are very different from the the number of the l_1 -embeddable regular hexagonal map on the torus and the Klein bottle.

2 Labels of l_1 -graph and 5-gonal Inequality

Here we introduce a kind of edge label (*atom of edge*) for l_1 -graphs, which was introduced by Shpectorov in [13]. Let G be a finite l_1 -graph and φ be a λ -embedding of G in Q_n . Assign each edge uv of G a label $l(uv)$ as follows: $l(uv) = \varphi(u) \triangle \varphi(v)$. For each edge $e = uv$ of G , $|\varphi(u) \triangle \varphi(v)| = d_{Q_n}(\varphi(u), \varphi(v)) = \lambda d_G(u, v) = \lambda$. Thus the label of each edge contains exactly λ elements from the set $\{1, 2, \dots, n\}$.

In order to simplify the notation, let Q_n represent the n -dimensional base set and the entire hypercube. Correspondingly, let uppercase letters represent the vertices of the hypercube, with $X \subset Q_n$ denoting a vertex. Suppose $x \mapsto X$ is a scale embedding of graph G into the hypercube Q_n . Then, by definition

$$d_{Q_n(X,Y)} = |X \triangle Y| = \lambda d(x, y),$$

where $x, y \in V(G)$, λ is the scale. At the same time, let v denote the vertex mapped to the empty set, i.e. $V = \emptyset$. Then we have the following lemma.

Lemma 2.1. [13] *Let λ be the scale of the embedding $x \mapsto X$ and $v \mapsto \emptyset$. Then:*

- (1) *For any $x \in G$ one has that $|X| = \lambda d(x, v)$ is an integer multiple of λ .*

(2) For any $x, y \in G$ one has that $|X \cap Y| = \frac{\lambda}{2}[d(x, v) + d(y, v) - d(x, y)]$ is an integer multiple of $\frac{\lambda}{2}$.

Lemma 2.2. [13] If $v = x_0, x_1, \dots, x_s = x$ is a shortest path from v to x , then X is the disjoint union of the proper atoms $A_i = X_i \setminus X_{i-1}, i = 1, \dots, s$.

Lemma 2.3. [16] For a simple l_1 -graph G , the labels on incident edges of G are never equal.

Lemma 2.4. [7, 10] Let u, v be two vertices of an l_1 -graph G and φ a λ -embedding of G into a hypercube. The following statements hold:

- (1) If $\rho = uu_1u_2u_3 \dots u_{k-1}v$ is a path from u to v , then $\varphi(u) \triangle \varphi(v) = l(uu_1) \triangle l(u_1u_2) \triangle l(u_2u_3) \triangle \dots \triangle l(u_{k-1}v)$.
- (2) If ρ is a shortest (u, v) -path, then the labels $l(uu_1), l(u_1u_2), l(u_2u_3), \dots, l(u_{k-1}v)$ are pairwise disjoint and $\varphi(u) \triangle \varphi(v) = l(uu_1) \cup l(u_1u_2) \cup l(u_2u_3) \cup \dots \cup l(u_{k-1}v)$.

Suppose that $C_k = v_0v_1v_2 \dots v_{k-1}v_0$ is a cycle of graph G . Two edges v_iv_{i+1} and v_jv_{j+1} ($0 \leq i, j \leq k-1$) are opposite if $d_{C_k}(v_i, v_j) = d_{C_k}(v_j, v_{j+1})$, and this distance is equal to the diameter of C_k . When $v_k = v_0$, it is easy to see that every edge has a unique opposite edge if k is even, and has two opposite edges if k is odd.

Lemma 2.5. [5, 9] Suppose that C_k is an isometric cycle of G and ab and cd are a pair of opposite edges of C_k . If k is even, then $l(ab) = l(cd)$, while if k is odd, then $|l(ab) \cap l(cd)| = \frac{\lambda}{2}$. Furthermore if k is odd and ce is the second edge opposite to ab then $l(ab) \subset l(cd) \cup l(ce)$. The labels of non-opposite edges are disjoint.

As documented in [14], there exists a necessary condition to characterize the l_1 -embeddable graphs. We summarize this assertion into the following lemma, which is frequently utilized to demonstrate that a graph is not l_1 -embeddable.

Lemma 2.6. [14] If G is an l_1 -graph, then for any five vertices x, y, a, b, c of G , they must satisfy the 5-gonal inequality:

$$d(x, y) + (d(a, b) + d(a, c) + d(b, c)) \leq (d(x, a) + d(x, b) + d(x, c)) + (d(y, a) + d(y, b) + d(y, c)).$$

3 l_1 -embeddability of 4-Regular Quadrilateral Maps on the Torus

In this section, we study of l_1 -embeddability of 4-regular quadrilateral maps on the torus.

According to [11], the two classes $Q_{k,m,r}$ and $Q_{k,m,e}$ of 4-regular quadrilateral maps on the torus are constructed as follows.

Let $P_k \square P_m$ be the Cartesian product of two paths P_k and P_m with $k, m \geq 1$. For convenience, support $P_k \square P_m$ is a grid with k vertical paths and m horizontal paths (see Figure 1). Assign to each vertex of $P_k \square P_m$ as follows: Denote the k vertical paths as: $R_1 = a_{1,0}a_{2,0} \dots a_{m,0}$, $R_2 = a_{1,1}a_{2,1} \dots a_{m,1}$, $\dots$, $R_k = a_{1,k-1}a_{2,k-1} \dots a_{m,k-1}$ and the m horizontal paths as: $T_1 = a_{1,0}a_{1,1} \dots a_{1,k-1}$, $T_2 = a_{2,0}a_{2,1} \dots a_{2,k-1}$, $\dots$, $T_m = a_{m,0}a_{m,1} \dots a_{m,k-1}$.

Let r be a natural number ($0 \leq r < k$). The map $Q_{k,m,r}$ is obtained from the grid graph $P_k \square P_m$ by adding all edges $a_{i,0}a_{i,k-1}$ ($i = 1, 2, \dots, m$) and $a_{1,i}a_{m,i+r}$ ($i = 0, 1, 2, \dots, k-1$; $i+r$ is module k). In particular, if $r = 0$, then $Q_{k,m,0} \cong C_k \square C_m$.

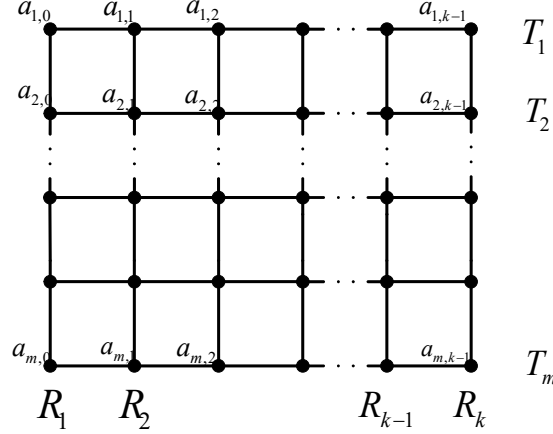


Figure 1: $P_k \square P_m$

It is easy to see that for any $0 \leq r < k$, $Q_{k,m,r} \cong Q_{k,m,k-r}$. In fact, the mapping $f : V(Q_{k,m,r}) \rightarrow V(Q_{k,m,k-r})$ defined by $f(a_{n,i}) = a_{n,(k-i-1)}$ ($n = 1, 2, \dots, m$ and $i = 0, 1, 2, \dots, k-1$) is an isomorphism between $Q_{k,m,r}$ and $Q_{k,m,k-r}$.

If $k = 1, m \geq 1$, then $r = 0$. In this case, the resulting graph $Q_{1,m,0}$, ignoring loops and multiple edges, is a cycle of length m . Therefore, it is an l_1 -graph.

If $k = 2, m = 1$, then $r = 0$ or 1 . The graphs $Q_{2,1,0}$ or $Q_{2,1,1}$, after deleting loops and multiple edges, is a path of length 1. Therefore, both $Q_{2,1,0}$ and $Q_{2,1,1}$ are l_1 -graphs.

If $k = 2, m = 2$, then $r = 0$ or 1 . When $r = 0$, the graph $Q_{2,2,0} \cong C_4$. When $r = 1$, the graph $Q_{2,2,1}$ is the complete graph K_4 . Clearly, both of them are l_1 -graphs.

Let C_k denote the cycle graph on k vertices with vertex set $\{0, 1, 2, \dots, k-1\}$. The map $Q_{k,m,e}$ is obtained from C_k by adding edges joining vertex i to vertex $(i+m)$, $0 \leq i \leq k-1$, where the addition is taken modulo k . We regard the added edges as undirected, so pairs of the form $\{i, i+m\}$ are taken in $\mathbb{Z}_k$. In fact, the map $Q_{k,m,e}$ is isomorphic to $\text{Circ}(k, \{\pm 1, \pm m\})$.

It is easy to see that $Q_{k,m,e} \cong Q_{k,k-m,e}$. Hence, it suffices to consider $Q_{k,m,e}$ with $k \geq 3$ and $1 \leq m \leq \lfloor \frac{k}{2} \rfloor$.

In order to more clearly represent the allowable range of the parameter r , we introduce a specific notation. Let

$$\mathcal{Q}_{k,m,r} = \{Q_{k,m,r} : k \geq 2, m \geq 2, 0 \leq r \leq \lfloor \frac{k}{2} \rfloor\},$$

and

$$\mathcal{Q}_{k,m,e} = \{Q_{k,m,e} : k \geq 3, 1 \leq m \leq \lfloor \frac{k}{2} \rfloor\}.$$

Reference [13] provides a result on the l_1 -embeddability of the Cartesian product of graphs.

Lemma 3.1. [13] *The Cartesian product of any two l_1 -graphs is also an l_1 -graph.*

Proposition 3.2. *If $G \in \mathcal{Q}_{k,m,0}$, then G is l_1 -embeddable.*

Proof. For any graph $G \in \mathcal{Q}_{k,m,0}$, it is known that G is isomorphic to the graph $C_k \square C_m$. By Lemma 3.1, G is an l_1 -graph. $\square$

Denote by $(k; a, b; p_a, p_b)$ and call *bifaced polyhedra* any k -regular polyhedron, whose faces are only p_a a -gons and p_b b -gons, where $3 \leq a \leq b$ and $p_a > 0$ and $p_b \geq 0$. By Euler's fomular, if $k = 4, a = 3$ and $p_b = 2$, then $p_3 = 2b$. For the bifaced polyhedra $(4; 3, b; 2b, 2)$, Deza and Grishukhin [3] denoted it $APrism_b$, and they proved the following proposition.

Proposition 3.3. [3] *Any bifaced polyhedra $APrism_b$ admits a 2-embedding into the Q_{b+1} .*

Proposition 3.4. *If $G \in \mathcal{Q}_{k,2,1}$, then G is l_1 -embeddable.*

Proof. Suppose that $G \in \mathcal{Q}_{k,2,1}$. If $k = 2$, then $Q_{2,2,1} \cong K_4$, which implies that $Q_{2,2,1}$ is an l_1 -graph. Now consider the graph $G \in \mathcal{Q}_{k,2,1}$ with $k \geq 3$. In fact, G is isomorphic to the 4-regular bifaced polyhedra of the form $(4; 3, k; 2k, 2)$. By Proposition 3.3, $(4; 3, k; 2k, 2)$ admits a 2-embedding into the hypercube Q_{k+1} . Therefore, G is l_1 -embeddable. $\square$

For convenience, we assume that the 4-regular quadrilateral maps $Q_{k,m,r}$ on the torus inherit the labeling of vertical/horizontal paths from the grid graph $P_k \square P_m$, denoted by T_i and R_i , respectively. When $k, m \geq 2$, it is easy to observe that the face cycle is isometric in G .

Moreover, each face cycle of the form $a_{1,i}a_{1,(i+1)}a_{m,(i+2)}a_{m,(i+1)}a_{1,i}$ (for $0 \leq i \leq k-1$, with indices taken modulo k) is also an isometric cycle (see Figure 2).

Theorem 3.1. *Let G be in $\mathcal{Q}_{k,m,r}$. Then G is an l_1 -graph if and only if G is in $\mathcal{Q}_{k,m,0} \cup \mathcal{Q}_{k,2,1}$.*

Proof. By Propositions 3.2 and 3.4, we know that if $G \in \mathcal{Q}_{k,m,0} \cup \mathcal{Q}_{k,2,1}$, then G is an l_1 -graph. Suppose that $G \in \mathcal{Q}_{k,m,r} \setminus \{\mathcal{Q}_{k,m,0} \cup \mathcal{Q}_{k,2,1}\}$, we will prove that G is not an l_1 -graph.

Suppose to the contrary that G is an l_1 -graph and it admits a λ -embedding ϕ into a hypercube Q_s for integers λ and s .

Case 1. $r = 1$.

Since the case $k \geq 2, m = 2$, and $r = 1$ means that $G \in \mathcal{Q}_{k,2,1}$, by Proposition 3.4, G is l_1 -embeddable. We now consider the case $k \geq 2, m \geq 3$, and $r = 1$.

Suppose that $G \in \mathcal{Q}_{k,m,r}$ with $k \geq 2, m \geq 3$, and $r = 1$ (See Figure 2), and it is an l_1 -graph. Then by Lemma 2.5, we obtain $l(a_{1,0}a_{1,1}) = l(a_{m,1}a_{m,2}) = l(a_{(m-1),1}a_{(m-1),2}) = \dots = l(a_{1,1}a_{1,2})$, which contradicts Lemma 2.3.

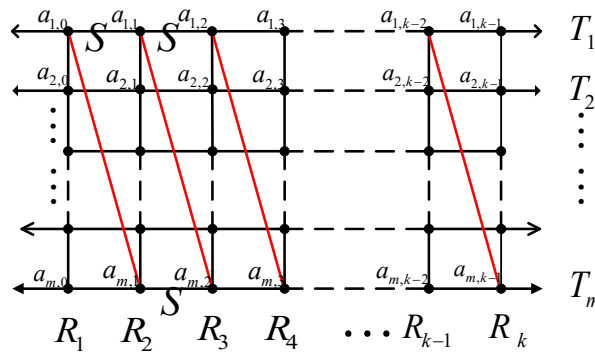


Figure 2: $G \in \mathcal{Q}_{k,m,r}$ (where $k \geq 2, m \geq 3, r = 1$)

Case 2. $2 \leq r \leq \lfloor \frac{k}{2} \rfloor$.

Take the cycle $C = a_{1,0}a_{1,1}a_{1,2} \dots a_{1,r}a_{2,r} \dots a_{m,r}a_{1,0}$ (see Figure 3). Since $k \geq 2, m \geq 2$, it is easy to see that the cycle C is an isometric cycle (see Figure 3).

Moreover, each face cycle of G is an isometric cycle, by Lemma 2.5, we have :

$$l(a_{1,0}a_{1,1}) = l(a_{2,0}a_{2,1}) = \cdots = l(a_{m,0}a_{m,1})$$

and

$$\begin{aligned} l(a_{1,0}a_{2,0}) &= l(a_{1,1}a_{2,1}) = \cdots = l(a_{1,r}a_{2,r}) \\ l(a_{2,0}a_{3,0}) &= l(a_{2,1}a_{3,1}) = \cdots = l(a_{2,r}a_{3,r}) \end{aligned}$$

$\vdots$

$$l(a_{m-1,0}a_{m,0}) = l(a_{m-1,1}a_{m,1}) = \cdots = l(a_{m-1,r}a_{m,r})$$

By Lemma 2.3, we have :

$$\begin{aligned} l(a_{1,0}a_{1,1}) \cap l(a_{1,1}a_{2,1}) &= \emptyset, \text{ i.e., } l(a_{1,0}a_{1,1}) \cap l(a_{1,r}a_{2,r}) = \emptyset. \\ l(a_{2,0}a_{2,1}) \cap l(a_{2,1}a_{3,1}) &= \emptyset, \text{ i.e., } l(a_{1,0}a_{1,1}) \cap l(a_{2,r}a_{3,r}) = \emptyset. \end{aligned}$$

$\vdots$

$$l(a_{m-1,0}a_{m-1,1}) \cap l(a_{m-1,1}a_{m,1}) = \emptyset, \text{ i.e., } l(a_{1,0}a_{1,1}) \cap l(a_{m-1,r}a_{m,r}) = \emptyset.$$

In, summary, we have :

$$l(a_{1,0}a_{1,1}) \cap l(a_{j,r}a_{j+1,r}) = \emptyset \text{ with } 1 \leq j \leq n-1. \quad (3.1)$$

Similarly, we have :

$$l(a_{1,r}a_{2,r}) \cap l(a_{1,i}a_{1,i+1}) = \emptyset \text{ with } 0 \leq i \leq r-1. \quad (3.2)$$

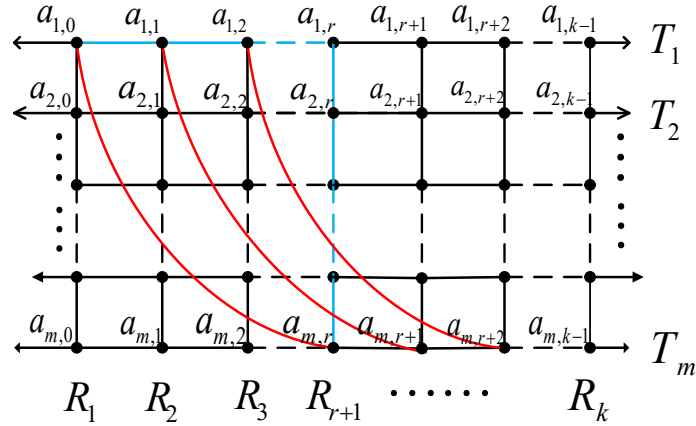


Figure 3: $G \in \mathcal{Q}_{k,m,r}$ (where $2 \leq k, 2 \leq m, 2 \leq r \leq \lfloor \frac{k}{2} \rfloor$)

Next, we will analyze the problem by considering the parity of the cycle C and comparing n with r .

If C is an even cycle and $m \geq r$, then we know that the edge opposite $a_{1,0}a_{1,1}$ must lie on R_{r+1} . Therefore, by Lemma 2.5, we have $|l(a_{1,0}a_{1,1}) \cap l(a_{j,r}a_{j+1,r})| = \lambda$ with $1 \leq j \leq m-1$. This is in contradiction with equation (3.1).

If C is an even cycle and $m < r$, then we know that the edge opposite $a_{1,r}a_{2,r}$ must lie on T_1 . Therefore, by Lemma 2.5, we have $|l(a_{1,r}a_{2,r}) \cap l(a_{1,i}a_{1,i+1})| = \lambda$ with $0 \leq i \leq r-1$. This is in contradiction with equation (3.2).

In the case where C is an odd cycle, and either $m \geq r$ or $m < r$, a similar argument applies. By Lemma 2.5, $|l(a_{1,0}a_{1,1}) \cap l(a_{j,r}a_{j+1,r})| = \frac{\lambda}{2}$ or $|l(a_{1,r}a_{2,r}) \cap l(a_{1,i}a_{1,i+1})| = \frac{\lambda}{2}$. This is in contradiction with equations (3.1) or (3.2).

Hence $G \in \mathcal{Q}_{k,m,r} \setminus \{\mathcal{Q}_{k,m,0} \cup \mathcal{Q}_{k,2,1}\}$ is not an l_1 -graph. $\square$

Here, we briefly recall how the second type of quadrilateral map $Q_{k,m,e}$ on the torus is constructed. Let C_k denote the cycle graph on $\mathbb{Z}_k$ and edge set $\{i(i+1) | i \in \mathbb{Z}_k\}$. The map $Q_{k,m,e}$ obtained from C_k by adding all edges joining vertex i to vertex $(i+m)$ for $0 \leq i \leq k-1$, where the addition is taken modulo k .

Lemma 3.5. *The map $Q_{2i,2,e}$ with $i \geq 2$ is l_1 -embeddable.*

Proof. Suppose that the vertex set of $Q_{2i,2,e}$ is $\{0, 1, \dots, 2i-1\}$. By the definition of $Q_{k,m,e}$, we know that the vertex i of $Q_{k,m,e}$ is adjacent to $i-1, i+1, i-m, i+m$. Hence in $Q_{2i,2,e}$, the vertex j is adjacent to $j-1, j+1, j-2$ and $j+2$.

Now define a mapping

$$f : V(Q_{2i,2,e}) \rightarrow V(Q_{i,2,1}),$$

$$f(j) = a_{2, \frac{j}{2}}, \text{ if } j \text{ is even or zero, and } f(j) = a_{1, \frac{j-1}{2}}, \text{ if } j \text{ is odd.}$$

It can be easily checked that the mapping f is an isomorphism between $Q_{2i,2,e}$ and $Q_{i,2,1}$. By Proposition 3.4, $Q_{i,2,1}$ is an l_1 -graph. Therefore $Q_{2i,2,e}$ is l_1 -embeddable. $\square$

Let

$$\mathcal{Q}_{k,m,e} = \{Q_{k,m,e} : k \geq 3, \left\lfloor \frac{k}{2} \right\rfloor \geq m \geq 1\}.$$

Proposition 3.6. *If $G \in \mathcal{Q}_{k,1,e}$, or $G \cong Q_{5,2,e}$, then G is l_1 -embeddable.*

Proof. Essentially, for any k , the graph G in $\mathcal{Q}_{k,1,e}$ is a cycle of length k (after deleting the multiple edges). If $m = 2$, then $k \geq 4$. The graph $Q_{5,2,e}$ is the complete graph on 5 vertices K_5 . So if $G \in \mathcal{Q}_{k,1,e}$, or $G \cong Q_{5,2,e}$, then G is an l_1 -graph. $\square$

Proposition 3.7. *$Q_{7,2,e}$ is not l_1 -embeddable.*

Proof. Consider the distances between the five vertices $x = 0, y = 3, a = 1, b = 2, c = 5$.

$$d_{G_i}(0, 3) + d_{G_i}(1, 2) + d_{G_i}(1, 5) + d_{G_i}(2, 5) = 7,$$

$$(d_{G_i}(0, 1) + d_{G_i}(0, 2) + d_{G_i}(0, 5)) + (d_{G_i}(3, 1) + d_{G_i}(3, 2) + d_{G_i}(3, 5)) = 6.$$

Which indicates that the five vertices $0, 1, 2, 3, 5$ of $Q_{7,2,e}$ violate the 5-gonal inequality, by Lemma 2.6, $Q_{7,2,e}$ is not l_1 -embeddable. $\square$

Proposition 3.8. *$Q_{9,2,e}$ is not l_1 -embeddable.*

Proof. The circulant graph $Q_{9,2,e}$ is showned in Figure 4. Suppose to the contrary that $Q_{9,2,e}$ is l_1 -embeddable and it admits a λ -embedding φ into a hypercube Q_s for some integers λ and s . Thus the label of each edge contains exactly λ elements of the set $\{1, 2, \dots, s\}$. Assign the vertex $\varphi(0)$ to represent the empty set $\emptyset$, then we have $l(01) = \varphi(0) \triangle \varphi(1) = \emptyset \triangle \varphi(1) = \varphi(1)$. Since 0 is adjacent to 1, by Lemma 2.1, we have $|\varphi(1)| = |l(01)| = \lambda$.

Similarly, we have

$$\begin{aligned} l(02) = \varphi(2), l(07) = \varphi(7), l(08) = \varphi(8), \text{ and} \\ |\varphi(1)| = |\varphi(2)| = |\varphi(7)| = |\varphi(8)| = \lambda. \end{aligned} \quad (3.3)$$

Since the vertices 0, 1, and 2 form a triangle, according to Lemma 2.5,

$$|l(01) \cap l(02)| = |\varphi(1) \cap \varphi(2)| = \frac{\lambda}{2}. \quad (3.4)$$

Then the label of $\varphi(0)$, $\varphi(1)$ and $\varphi(2)$ consist as follows:

$$\begin{aligned} \varphi(0) &= \emptyset, \\ \varphi(1) &= (\varphi(1) \setminus \varphi(2)) + (\varphi(1) \cap \varphi(2)), \\ \varphi(2) &= (\varphi(2) \setminus \varphi(1)) + (\varphi(1) \cap \varphi(2)). \end{aligned} \quad (3.5)$$

One hand, the vertices 0, 1, 8 and 0, 7, 8 forms a triangle, respectively. Then

$$\begin{aligned} |l(01) \cap l(08)| &= |\varphi(1) \cap \varphi(8)| = \frac{\lambda}{2}, \\ |l(08) \cap l(07)| &= |\varphi(8) \cap \varphi(7)| = \frac{\lambda}{2}. \end{aligned} \quad (3.6)$$

Since the paths 107, 207 and 208 are the shortest 1, 7-path, 2, 7-path and 2, 8-path, by Lemma 2.4,

$$\begin{aligned} l(01) \cap l(07) &= \varphi(1) \cap \varphi(7) = \emptyset, \\ l(02) \cap l(07) &= \varphi(2) \cap \varphi(7) = \emptyset, \\ l(02) \cap l(08) &= \varphi(2) \cap \varphi(8) = \emptyset. \end{aligned} \quad (3.7)$$

By contrasting the equations (3.6) and (3.7), we can obtain that the λ elements of $\varphi(8)$ are consist of $\frac{\lambda}{2}$ elements of $\varphi(1) \cap \varphi(8)$ and the other $\frac{\lambda}{2}$ elements of $\varphi(8) \cap \varphi(7)$. While $\varphi(1) \cap \varphi(7) = \emptyset$, so

$$\varphi(8) = (\varphi(1) \cap \varphi(8)) + (\varphi(8) \cap \varphi(7)),$$

here the “+” represents the disjoint union of two sets, and the set addition in the subsequent context carries the same meaning.

By the equation (3.7), $\varphi(2) \cap \varphi(8) = \emptyset$. Then the $\frac{\lambda}{2}$ elements of $\varphi(1) \cap \varphi(8)$ must come from $\varphi(1) \setminus \varphi(2)$. As known from the equations (3.3) and (3.4), $|\varphi(1)| = \lambda$, $|\varphi(1) \cap \varphi(2)| = \frac{\lambda}{2}$, we get $|\varphi(1) \setminus \varphi(2)| = \frac{\lambda}{2}$. Therefore,

$$\varphi(1) \cap \varphi(8) = \varphi(1) \setminus \varphi(2). \quad (3.8)$$

On the other hand, the path 013 is the shortest 0, 3-path, by Lemma 2.4 and the distance formula in the hypercube,

$$|\varphi(3) \triangle \varphi(0)| = |\varphi(3) \triangle \emptyset| = |\varphi(3)| = d_{Q_n}(\varphi(0), \varphi(3)) = \lambda \cdot d_G(0, 3) = 2\lambda. \quad (3.9)$$

By Lemma 2.2, $\varphi(1) \subset \varphi(3)$. Similarly, since the path 023 is the shortest 0, 3-path, $\varphi(2) \subset \varphi(3)$. From the figure 4, it is easy to see that the cycle 023570 is an isometric 5-cycle, and the edges 23 and 07 are opposite. By Lemma 2.5,

$$|l(23) \cap l(07)| = |(\varphi(2) \triangle \varphi(3)) \cap \varphi(7)| = |\varphi(3) \cap \varphi(7)| = \frac{\lambda}{2}. \quad (3.10)$$

From the equation (3.7), the elements of $\varphi(7)$ are completely distinct from those of $\varphi(1)$ and $\varphi(2)$. Then the elements of $\varphi(3)$ can be written as

$$\begin{aligned}\varphi(3) &= (\varphi(1) \setminus \varphi(2)) + (\varphi(2) \setminus \varphi(1)) + (\varphi(1) \cap \varphi(2)) + (\varphi(7) \setminus \varphi(3)) \\ &= (\varphi(1) \setminus \varphi(2)) + \varphi(2) + (\varphi(7) \setminus \varphi(3))\end{aligned}\quad (3.11)$$

Similarly, the paths 086 and 076 are two shortest 0, 6-paths, respectively. Then

$$|\varphi(6)| = 2\lambda, \varphi(8) \subset \varphi(6), \varphi(7) \subset \varphi(6). \quad (3.12)$$

Since the cycle 024680 is an isometric 5-cycle, and the edges 02 and 68 are opposite,

$$|l(02) \cap l(68)| = |\varphi(2) \cap (\varphi(6) \cap \varphi(8))| = |\varphi(2) \cap \varphi(6)| = \frac{\lambda}{2}. \quad (3.13)$$

Then the elements of $\varphi(6)$ can be written as

$$\begin{aligned}\varphi(6) &= \varphi(7) \cup \varphi(8) \cup (\varphi(2) \cap \varphi(6)) \\ &= \varphi(7) \cup \left((\varphi(1) \cap \varphi(8)) \cup (\varphi(8) \cap \varphi(7)) \right) \cup (\varphi(2) \cap \varphi(6)) \\ &= \varphi(7) + (\varphi(1) \cap \varphi(8)) + (\varphi(2) \cap \varphi(6)).\end{aligned}\quad (3.14)$$

By the equation (3.8), $\varphi(1) \cap \varphi(8) = \varphi(1) \setminus \varphi(2)$. Then the $\varphi(3) \triangle \varphi(6) = (\varphi(7) \setminus \varphi(3)) \cup (\varphi(2) \setminus \varphi(6))$. As above shown, $\varphi(2) \cap \varphi(7) = \emptyset$, $|\varphi(7)| = \lambda$, $|\varphi(7) \cap \varphi(3)| = \frac{\lambda}{2}$, so $|\varphi(7) \setminus \varphi(3)| = \frac{\lambda}{2}$. Since $|\varphi(2)| = \lambda$, $|\varphi(2) \cap \varphi(6)| = \frac{\lambda}{2}$, $|\varphi(2) \setminus \varphi(6)| = \frac{\lambda}{2}$.

Therefore,

$$d_{Q_n}(\varphi(3), \varphi(6)) = |\varphi(3) \triangle \varphi(6)| \stackrel{\text{by (3.11) and (3.14)}}{=} |(\varphi(6) \setminus \varphi(3)) \cup (\varphi(2) \setminus \varphi(6))| = \lambda. \quad (3.15)$$

This means that the vertex 3 is adjacent to the vertex 6 in $Q_{9,2,e}$, which is a contradiction. $\square$

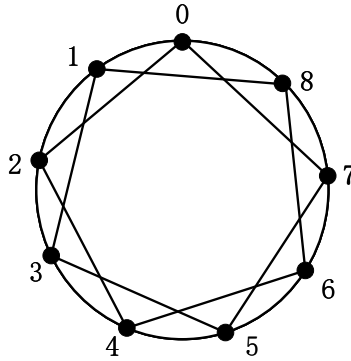


Figure 4: $Q_{9,2,e}$

Theorem 3.2. *For every odd integer $k \geq 11$, $Q_{k,2,e}$ is not an l_1 -graph.*

Proof. Suppose that $k = 2i + 1$, then $i \geq 5$. Denote $Q_{k,2,e}$ as G_i . We construct five vertices that violate Lemma 2.6. The construction depends on the parity of i .

Claim 1. For all $a, b \in \mathbb{Z}_{2i+1}$,

$$d_{G_i}(a, b) = \left\lceil \frac{\rho(a, b)}{2} \right\rceil. \quad (3.16)$$

Each edge of G_i changes the cyclic position by 1 or 2, so one step can reduce the cyclic distance by at most 2. This gives the lower bound

$$d_{G_i}(a, b) \geq \left\lceil \frac{\rho(a, b)}{2} \right\rceil.$$

Conversely, along a shorter arc of length $\rho(a, b)$ in the underlying cycle, one reaches b from a by using jumps of length 2, together with one jump of length 1 if the arc length is odd. This realizes exactly $\lceil \rho(a, b)/2 \rceil$ steps. Hence (3.16) follows.

Case 1: i is odd

Let $i = 2q + 1$, $q \geq 2$. Since $i > 4$, this covers all odd $i \geq 5$. In $\mathbb{Z}_{2i+1}$, choose $a = 1$, $b = 2$, $c = i + 2$, and $x = 0, y = 3$.

Here $2i + 1 = 4q + 3$ and $c = i + 2 = 2q + 3$. By Claim 1,

$$d(a, b) = d(1, 2) = 1, d(a, c) = d(1, 2q + 3) = q + 1, d(b, c) = d(2, 2q + 3) = q + 1.$$

and

$$d(x, y) = d(0, 3) = 2.$$

Thus the left-hand side of the pentagonal inequality equals

$$L = 1 + (q + 1) + (q + 1) + 2 = 2q + 5 = i + 4. \quad (3.17)$$

On the other hand,

$$d(0, 1) = 1, \quad d(0, 2) = 1, \quad d(0, i + 2) = d(0, 2q + 3) = q;$$

and

$$d(3, 1) = 1, \quad d(3, 2) = 1, \quad d(3, i + 2) = d(3, 2q + 3) = q.$$

Therefore the right-hand side of Lemma 2.6 equals

$$R = 1 + 1 + 1 + 1 + q + q = 2q + 4 = i + 3. \quad (3.18)$$

Equations (3.17) and (3.18) give

$$L = i + 4 > i + 3 = R,$$

which contradicts the pentagonal inequality.

Case 2: i is even

Let $i = 2q$, $q \geq 3$. Since $i > 4$, this covers all even $i \geq 6$. In $\mathbb{Z}_{2i+1}$, choose $a = 1, b = 4, c = i + 3$, and $x = 0, y = 5$.

Here $2i + 1 = 4q + 1$ and $c = i + 3 = 2q + 3$. By Claim 1,

$$d(a, b) = d(1, 4) = 2, d(a, c) = d(1, 2q + 3) = q, d(b, c) = d(4, 2q + 3) = q, d(x, y) = d(0, 5) = 3.$$

Hence the left-hand side of Lemma 2.6 is

$$L = 2 + q + q + 3 = 2q + 5 = i + 5. \quad (3.19)$$

The six distances of the right-hand side of Lemma 2.6 are

$$d(1, 0) = 1, d(0, 4) = 2, d(0, i + 3) = d(0, 2q + 3) = q - 1,$$

and

$$d(5, 1) = 2, d(5, 4) = 1, d(5, i + 3) = d(5, 2q + 3) = q - 1.$$

Thus the right-hand side of Lemma 2.6 is

$$R = 1 + 2 + 2 + 1 + (q - 1) + (q - 1) = 2q + 4 = i + 4. \quad (3.20)$$

Equations (3.19) and (3.20) give

$$L = i + 5 > i + 4 = R,$$

again violating the pentagonal inequality.

In both parity cases, five vertices of G_i violate the pentagonal inequality required of ℓ_1 metrics. Therefore the graph $Q_{k,2,e}$ with odd $k \geq 11$ cannot be ℓ_1 -embeddable. $\square$

Theorem 3.3. *Let G be in $\mathcal{Q}_{k,m,e}$. Then G is an l_1 -graph if and only if it is isomorphic to $Q_{5,2,e}$, or belongs to $\mathcal{Q}_{k,1,e} \cup \mathcal{Q}_{2i,2,e}$, where $2 \leq i \in \mathbb{N}$.*

Proof. By Lemma 3.5 and Proposition 3.6, we know that if G is isomorphic to $Q_{5,2,e}$, or belongs to $\mathcal{Q}_{k,1,e} \cup \mathcal{Q}_{2i,2,e}$, where $2 \leq i \in \mathbb{N}$, then G is an l_1 -graph.

Next, we will prove that if $G \in \mathcal{Q}_{k,m,e} \setminus \{\mathcal{Q}_{k,1,e} \cup \mathcal{Q}_{2i,2,e} \cup \{Q_{5,2,e}\}\}$, then G is not an l_1 -graph.

Case 1. $m = 2$.

According to the choice of G , then just consider the odd number $k \neq 5$. So it suffices to consider $k \geq 7$. let $k = 2i + 1$ (i.e., $i \geq 3$).

If $i = 3, 4$, by Propositions 3.7 and 3.8, both $Q_{7,2,e}$ and $Q_{9,2,e}$ are not l_1 -embeddable.

If $i > 4$, then $k \geq 11$, by Theorem 3.2, $Q_{k,2,e}$ is not l_1 -embeddable.

Case 2. $3 \leq m < \left\lceil \frac{k}{2} \right\rceil$.

If $G \in \mathcal{Q}_{k,m,e}$ with $3 \leq m < \left\lceil \frac{k}{2} \right\rceil$, then $k \geq 8$. As illustrated in [?], part of $Q_{38,12,e}$ is shown in Figure 5. It is clear that each quadrilateral face in G is an isometric cycle.

Suppose to the contrary that G is l_1 -embeddable. Let the label of edge $\{0, m\}$ be η . By Lemma 2.5, $l(0, m) = l(1, m + 1) = l(2, m + 2) = \dots = l(m, 2m) = \eta$, which contradicts Lemma 2.3. Therefore, $G \in \mathcal{Q}_{k,m,e}$ with $3 \leq m < \left\lceil \frac{k}{2} \right\rceil$ is not an l_1 -graph.

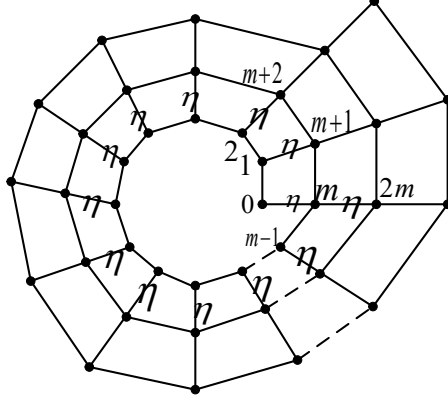


Figure 5: part of $Q_{k,m,e}$ with $3 \leq m < \left\lfloor \frac{k}{2} \right\rfloor$

Case 3. $m = \left\lfloor \frac{k}{2} \right\rfloor$.

If $G \in \mathcal{Q}_{k,m,e}$ with $m = \left\lfloor \frac{k}{2} \right\rfloor$, we consider the cases according to the parity of k .

Subcase 1. $k = 2i$ ($i \in \mathbb{N}$).

In Lemma 3.5, we have already shown that $Q_{4,2,e}$ is an l_1 -graph. Therefore, in the following discussion, we only consider the case $i \geq 3$. Since $i \equiv -i \pmod{2i}$, every vertex has a unique diametrical vertex in the graph $Q_{2i,i,e}$, and the degree of each vertex is 3. Denote $Q_{2i,i,e}$ as G_i .

First we show that for $0 \leq t \leq i$, $d_{G_i}(0, t) = \min\{t, i + 1 - t\}$. Along the outer cycle C_{2i} , one can go from 0 to t in t steps. On the other hand, one may use the chord $0 \sim i$ and then move from i to t along the cycle, which takes $1 + (i - t) = i + 1 - t$ steps. Hence

$$d_{G_i}(0, t) \leq \min\{t, i + 1 - t\}.$$

Conversely, every walk from 0 to t either avoids the chord joining antipodal vertices, in which case its length is at least t , or uses such a chord at least once, in which case after crossing to the opposite side one still needs at least $i - t$ steps along the cycle, so the total length is at least $1 + (i - t) = i + 1 - t$. Therefore

$$d_{G_i}(0, t) \geq \min\{t, i + 1 - t\},$$

and the formula $d_{G_i}(0, t) = \min\{t, i + 1 - t\}$ follows. Consequently, for arbitrary $a, b \in \mathbb{Z}_{2i}$, the distance $d_{G_i}(a, b)$ depends only on $|a - b|$ modulo $2i$.

Set

$$r := \left\lfloor \frac{i + 1}{2} \right\rfloor,$$

and choose five vertices of G_i by

$$a = 0, \quad b = r, \quad c = i + 1, \quad x = 1, \quad y = i.$$

We compute the distances appearing in the pentagonal inequality.

First, we have:

$$d(a, b) = d(0, r) = r, d(a, c) = d(0, i + 1) = 2,$$

since $0 \sim i \sim i + 1$; and

$$d(b, c) = d(r, i + 1) = \min\{i + 1 - r, r\} = r,$$

Also, $d(x, y) = d(1, i) = 2$, since $1 \sim 0 \sim i$. Therefore the left-hand side of the pentagonal inequality equals

$$L := d(a, b) + d(a, c) + d(b, c) + d(x, y) = r + 2 + r + 2 = 2r + 4.$$

Now we compute the six mixed distances. Clearly,

$$d(x, a) = d(1, 0) = 1, \quad d(x, b) = d(1, r) = r - 1, \quad d(x, c) = d(1, i + 1) = 1,$$

Moreover,

$$d(y, a) = d(i, 0) = 1, \quad d(y, b) = d(i, r) = \left\lceil \frac{i-1}{2} \right\rceil, \quad d(y, c) = d(i, i+1) = 1.$$

Hence the right-hand side of the pentagonal inequality equals

$$\begin{aligned} R &:= (d(x, a) + d(x, b) + d(x, c)) + (d(y, a) + d(y, b) + d(y, c)) \\ &= (d(1, 0) + d(1, r) + d(1, i+1)) + (d(i, 0) + d(i, r) + d(i, i+1)) \\ &= (1 + r - 1 + 1) + (1 + \left\lceil \frac{i-1}{2} \right\rceil + 1) \\ &= 3 + \left\lfloor \frac{i+1}{2} \right\rfloor + \left\lceil \frac{i-1}{2} \right\rceil. \end{aligned} \tag{3.21}$$

We now compare L and R .

If $i = 2q$ is even, then $r = \lfloor (2q+1)/2 \rfloor = q$, $\left\lfloor \frac{i+1}{2} \right\rfloor + \left\lceil \frac{i-1}{2} \right\rceil = 2q$, so

$$L = 2q + 4, \quad R = 4 + (q - 1) + q = 2q + 3.$$

Thus $L > R$.

If $i = 2q + 1$ is odd, then $r = \lfloor (2q+2)/2 \rfloor = q + 1$, so

$$L = 2(q + 1) + 4 = 2q + 6, \quad R = 3 + (q + 1) + q = 2q + 4.$$

Again $L > R$.

In both cases the five vertices $0, 1, \lfloor \frac{i+1}{2} \rfloor, i$ and $i+1$ violate the 5-gonal inequality in Lemma 2.6.

Hence, the graph G of $\mathcal{Q}_{2i, i, e}$ with $m = \lfloor \frac{k}{2} \rfloor = i$ and $k = 2i$ ($3 \leq i \in \mathbb{N}$) is not an l_1 -graph.

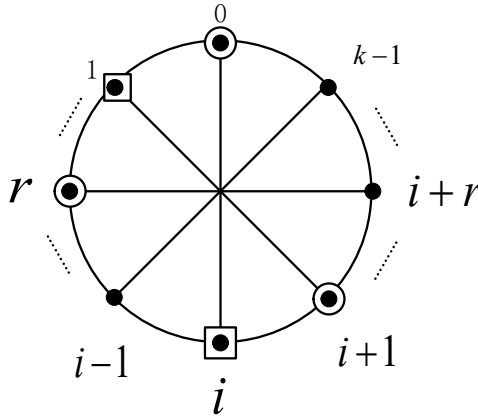


Figure 6: Graph $G \in \mathcal{Q}_{2i, m, e}$ with $m = \lfloor \frac{k}{2} \rfloor$

Subcase 2. $k = 2i + 1$ ($3 \leq i \in \mathbb{N}$).

By Proposition 3.6, $Q_{3,1,e}$ and $Q_{5,2,e}$ are l_1 -graphs. Now we only consider the case $k \geq 7$. Let's prove the following important claim.

Claim 2. *The graph $Q_{2i+1,m,e}$ is isomorphic to the graph $Q_{2i+1,2,e}$ for some $3 \leq i \in \mathbb{N}$.*

Define a mapping :

$$\tau : V(Q_{2i+1,m,e}) \rightarrow V(Q_{2i+1,2,e}) \text{ for } i \geq 3,$$

such that for $t \in \{0, 1, 2, \dots, k-1\}$

$$\tau(t) = 2t - 1 \text{ module } k.$$

Suppose that $u \in V(Q_{2i+1,m,e})$ whose neighbors are $u+1$, $u-1$, $u+m$ and $u-m$. According to the mapping τ , we have

$$\begin{aligned} \tau(u) &= 2u - 1, \tau(u-1) = 2u - 3, \tau(u+1) = 2u + 1, \tau(u+m) = 2u + 2m - 1, \\ \tau(u-m) &= 2u - 2m - 1. \end{aligned}$$

Since $m = \left\lfloor \frac{k}{2} \right\rfloor$, then we have $k = 2m + 1$. Therefore,

$$\tau(u+m) = 2u - 2 \text{ and } \tau(u-m) = 2u.$$

However, in the graph $Q_{2i+1,2,e}$, the vertex $2u - 1$ has neighbors $2u$, $2u - 2$, $2u + 1$, and $2u - 3$. That means that the adjacency relations in $Q_{2i+1,m,e}$ are preserved in $Q_{2i+1,2,e}$ under the mapping τ . Hence, τ is a graph isomorphism and $Q_{2i+1,m,e} \cong Q_{2i+1,2,e}$. The Claim 2 holds.

However, we have already proven in Theorem 3.3 that $Q_{2i+1,2,e}$ ($i \in \mathbb{N}$ with $i \geq 3$) is not an l_1 -graph. Therefore, graph $Q_{2i+1,m,e}$ with $m = \left\lfloor \frac{k}{2} \right\rfloor$ and $k = 2i + 1$ ($i \in \mathbb{N}$ with $i \geq 3$) is not an l_1 -graph. $\square$

From Propositions 3.2, 3.4, 3.6, 3.7, 3.8. Lemma 3.5 and Theorem 3.1, 3.2, 3.3, we can obtain the following main theorem.

Theorem 3.4. *Among the regular quadrilateral maps on the torus, only the maps $\mathcal{Q}_{k,m,0}$, $\mathcal{Q}_{k,2,1}$, $\mathcal{Q}_{k,1,e}$, $Q_{5,2,e}$ and $\mathcal{Q}_{2i,2,e}$ with $2 \leq i \in \mathbb{N}$ are l_1 -embeddable.*

Let n be an integer. Identify the vertex set of the *circulant graph* $\text{Circ}(n, \{\pm 1, \pm m\})$ with the cyclic group $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$. Two vertices $x, y \in \mathbb{Z}_n$ are adjacent if and only if $y - x = \pm 1, \pm m \pmod{n}$. According to the definition of the map $Q_{k,m,e}$ with $k \geq 3, m \geq 1$, it is isomorphic to the circulant graph $\text{Circ}(k, \{\pm 1, \pm m\})$. From Theorem 3.4, we can easily obtain that

Corollary 3.1. *Among the circulant graphs, only $\text{Circ}(5, \{\pm 1, \pm 2\})$ and $\text{Circ}(2i, \{\pm 1, \pm 2\})$ ($i \in \mathbb{Z}^+$) are l_1 -embeddable.*

4 l_1 -embeddability of Regular Quadrilateral Maps on the Klein Bottle

In this section, we study the l_1 -embeddability of 4-regular quadrilateral maps on the Klein bottle.

As mentioned earlier in [11], there are six types of 4-regular quadrilateral maps $Q_{k,m,a}$, $Q_{k,m,b}$, $Q_{k,m,c}$, $Q_{k,m,f}$, $Q_{k,m,g}$, and $Q_{k,m,h}$ on the Klein bottle. However, it was shown in fact there are only three distinct types [20], since $Q_{k,m,c} \cong Q_{2m, \frac{k}{2}, a}$, $Q_{k,m,f} \cong Q_{2(m+1), \frac{k}{2}, b}$, $Q_{k,m,g} \cong Q_{2m+1, \frac{k}{2}, a}$.

Therefore, it is only necessary to discuss which of $Q_{k,m,a}$, $Q_{k,m,b}$ and $Q_{k,m,h}$ are l_1 -embeddable.

First, define the Cartesian product of a cycle C_k of length k and a path P_m on m vertices, denoted by $C_k \square P_m$, as a *quadrilateral cylinder of length k and width m* . The two k -cycles consisting of the vertices of degree 3 are referred to as the boundary cycles. Label the vertices on these two boundary cycle as $x_1, x_2, \dots, x_k, x_1$ and $y_1, y_2, \dots, y_k, y_1$, respectively, such that x_i and y_i are the two end-vertices of the same copy of P_m . The vertices of degree 4 are denoted by $u_{i,j}$, where $1 \leq i \leq m-2$ and $1 \leq j \leq k$ (see Figure 7). These vertices correspond precisely to the interior vertices of the quadrilateral cylinder, i.e., those not lying on the boundary cycles.

For integer $k \geq 2$ and $m \geq 2$, the map $Q_{k,m,a}$ is defined as a graph obtained from a quadrilateral cylinder of length k and width m by adding the following set of edges along its boundary cycles:

$$x_1y_2, x_2y_1, x_ky_3, x_3y_k, x_{k-1}y_4, x_4y_{k-1}, \dots$$

When k is odd, the construction is illustrated in Figure 7b, and when k is even, it is illustrated in Figure 7a.

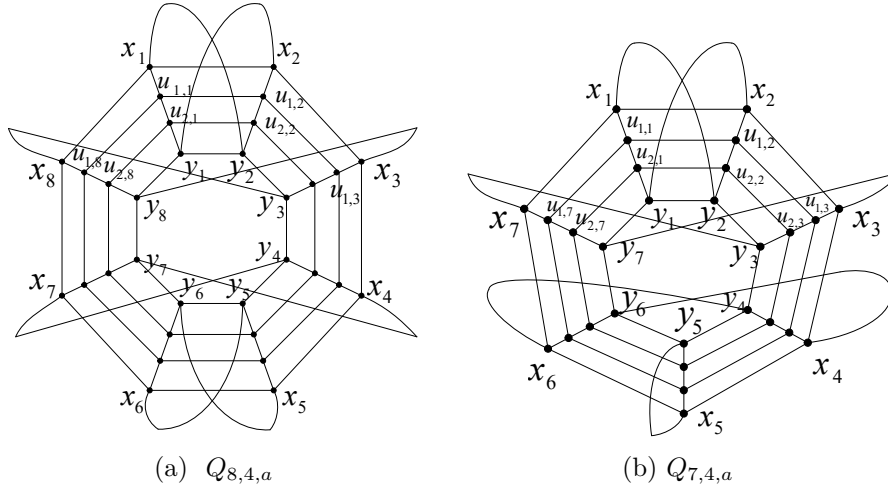


Figure 7: example of $Q_{k,m,a}$

Theorem 4.1. *The map $Q_{k,m,a}$ with $k \geq 2$ and $m \geq 2$ is l_1 -embeddable if and only if $k = 2, m = 2$ or $k = 4, m = 2$.*

Proof. For the graph $Q_{2,2,a}$, after removing loops and multiple edges, it is in fact isomorphic to K_4 . Hence, $Q_{2,2,a}$ is an l_1 -graph. The graph $Q_{4,2,a}$ can be 2-embedded into the hypercube Q_6 . The corresponding embedding of the vertices into the Q_6 is shown in Figure 8.

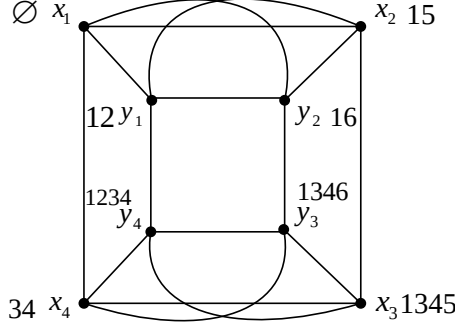


Figure 8: $Q_{4,2,a}$

Let $\mathcal{Q}_{k,m,a} = \{Q_{k,m,a} : k \geq 2, m \geq 2\}$.

In the following, we will show that any graph of $G \in \mathcal{Q}_{k,m,a} \setminus \{Q_{2,2,a}, Q_{4,2,a}\}$ is not l_1 -embeddable. Suppose that $Q_{k,m,a}$ is l_1 -embeddable, and it admits a λ -embedding ϕ into a hypercube Q_s for some integers λ and s .

Case 1. $k \geq 2$ and $m = 2$.

Since the cycle $x_1 x_k y_k y_1 x_1$ is isometric, then by Lemma 2.5, we have $l(x_1 x_k) = l(y_1 y_k)$. Furthermore, as the cycle $x_2 x_3 y_k y_1 x_2$ is also isometric, it follows that $l(x_2 x_3) = l(y_1 y_k)$. Hence, we obtain $l(x_2 x_3) = l(x_1 x_k)$.

Since $x_k x_1 x_2 x_3$ is a path from vertex x_k to x_3 , by Lemma 2.4, we have

$$|\phi(x_k) \Delta \phi(x_3)| = |l(x_k x_1) \Delta l(x_1 x_2) \Delta l(x_2 x_3)| = \lambda.$$

This implies that the vertices x_3 and x_k are adjacent in G , which leads to a contradiction.

Case 2. $k \geq 2$ and $m \geq 3$.

For the case $k \geq 2$ and $m \geq 3$, we observe that every face cycle of the graph $Q_{k,m,a}$ is an isometric cycle (see Figure 7). Therefore, we have

$$l(x_1 x_2) = l(u_{1,1} u_{1,2}) = l(u_{2,1} u_{2,2}) = \cdots = l(y_1 y_2).$$

Moreover, by Lemma 2.3, we have

$$\begin{aligned} l(x_1 x_2) \cap l(x_1 u_{1,1}) &= \emptyset, l(u_{1,1} u_{1,2}) \cap l(u_{1,1} u_{2,1}) = \emptyset, \\ l(u_{2,1} u_{2,2}) \cap l(u_{2,1} u_{3,1}) &= \emptyset, \dots, l(u_{m-2,1} u_{m-2,2}) \cap l(u_{m-2,1} y_1) = \emptyset. \end{aligned}$$

equivalent to

$$\begin{aligned} l(y_1 y_2) \cap l(x_1 u_{1,1}) &= \emptyset, l(y_1 y_2) \cap l(u_{1,1} u_{2,1}) = \emptyset, \\ l(y_1 y_2) \cap l(u_{2,1} u_{3,1}) &= \emptyset, \dots, l(y_1 y_2) \cap l(u_{m-2,1} y_1) = \emptyset. \end{aligned} \quad (4.1)$$

This implies that the label of $y_1 y_2$ does not intersect any of the labels of the edges $x_1 u_{1,1}$, $u_{1,1} u_{2,1}$, $u_{2,1} u_{3,1}$, $\dots$, $u_{m-2,1} y_1$.

From Figure 7, we observe that the cycle $C_1 : x_1 u_{1,1} u_{2,1} \dots y_1 y_2 x_1$ is an isometric cycle. Hence, regardless of whether it is an even cycle or an odd cycle, the edge $y_1 y_2$ must have one

opposite edge or two opposite edges in C_1 . According to Lemma 2.5, the intersection of the label of edge y_1y_2 with the label of its opposite edge has $\frac{\lambda}{2}$ or λ elements, which contradicts equation (4.1).

Therefore, $\mathcal{Q}_{k,m,a}$ with $k \geq 2$ and $m \geq 3$ is not l_1 -embeddable. $\square$

Now, we consider the l_1 -embeddability of the map $Q_{k,m,b}$ with $k \geq 2, m \geq 2$.

For even k , the map $Q_{k,m,b}$ is defined by adding the edges $x_1y_1, x_2y_k, x_ky_2, \dots, x_{\frac{k}{2}+1}y_{\frac{k}{2}+1}$ to the Cartesian product $C_k \square P_m$, as illustrated in Figure 9.

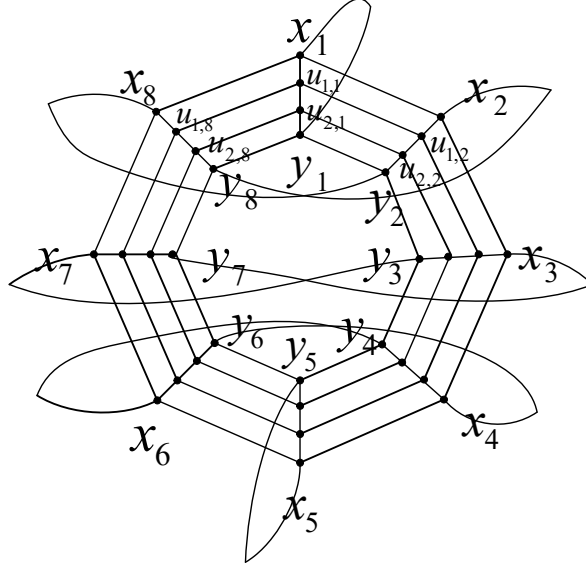


Figure 9: $Q_{8,4,b}$

For odd k , by adding edges in a manner analogous to the even case, the resulting map $Q_{k,m,b}$ is isomorphic to $Q_{k,m,a}$ [20]. Therefore, this case does not need to consider again.

Let $\mathcal{Q}_{k,m,b} = \{Q_{k,m,b} : k \geq 2 \text{ is even}, m \geq 2\}$.

Theorem 4.2. *Any map $Q_{k,m,b}$ of $\mathcal{Q}_{k,m,b}$ is l_1 -embeddable if and only if $k = 2$.*

Proof. When $k = 2$, the graph $Q_{2,m,b}$, after removing multiple edges, is isomorphic to $P_2 \square C_m$. By Lemma 3.1, $Q_{2,m,b}$ is l_1 -embeddable.

In the following, we show that if $k \neq 2$, then the map $Q_{k,m,b}$ does not admit an l_1 -embedding.

If $k \geq 4, m \geq 2$, from Figure 9, it is easy to see that every face cycle of $Q_{k,m,b}$ is isometric, by Lemma 2.5, which implies

$$l(x_1x_2) = l(u_{1,1}u_{1,2}) = l(u_{2,1}u_{2,2}) = \dots = l(y_1y_2).$$

Note that $u_{i,j}u_{i,j+1}$ ($1 \leq i \leq j \leq m-2$) may not exist.

Moreover, since the cycle $x_1x_2y_ky_1x_1$ is isometric, by Lemma 2.5, $l(x_1x_2) = l(y_1y_k)$. Hence, $l(y_1y_2) = l(y_1y_k)$. However, the edges y_1y_2 and y_1y_k are adjacent, which contradicts Lemma 2.3. So the map $\mathcal{Q}_{k,m,b}$ with $k \geq 4$ is not l_1 -embeddable. $\square$

Finally, we study the l_1 -embeddability of the quadrilateral map on the Klein bottle $Q_{k,m,h}$ [?].

For integers $k \geq 1$ and $m \geq 2$, the map $Q_{k,m,h}$ is defined as follows. First take the Cartesian product $P_m \square P_m$ of two paths on m vertices. Let $x_1 x_2 \dots x_m$, $y_1 y_2 \dots y_m$, $u_1 u_2 \dots u_m$, and $v_1 v_2 \dots v_m$ be the four paths in $P_m \square P_m$, such that

$$x_m = y_1, x_1 = u_m, v_1 = u_1, y_m = v_m.$$

The vertices y_1, x_1, v_1, y_m are of degree 2, while all other vertices x_i, y_i, u_i, v_i ($i = 2, 3, \dots, m-1$) are of degree 3.

Next, add m paths

$$x_1 z_{1,1} z_{2,1} \dots z_{k,1} y_1, x_2 z_{1,2} z_{2,2} \dots z_{k,2} y_2, \dots, x_m z_{1,m} z_{2,m} \dots z_{k,m} y_m,$$

where all newly introduced vertices $z_{i,j}$ ($i = 1, 2, \dots, k$, $j = 1, 2, \dots, m$) are distinct. Then we connect the vertices to form the paths $z_{i,1} z_{i,2} \dots z_{i,m}$, ($i = 1, 2, \dots, k$), as illustrated in Figure 10.

Finally, add the edges

$$u_1 z_{1,m}, u_2 z_{2,m}, \dots, z_{k-1,1} v_2, z_{k,1} v_1,$$

the result graph is the *quadrilateral map on the Klein bottle* $Q_{k,m,h}$ (see Figure 11).

For convenience, label the vertices on the path from u_{m-1} to y_2 as $w_1, w_2, \dots, w_m$, where $w_1 = u_{m-1}$ and $w_m = y_2$ (see Figure 10).

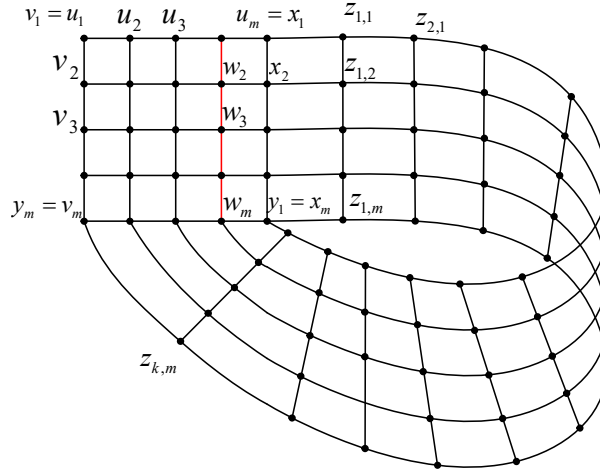


Figure 10: part of $Q_{k,m,h}$

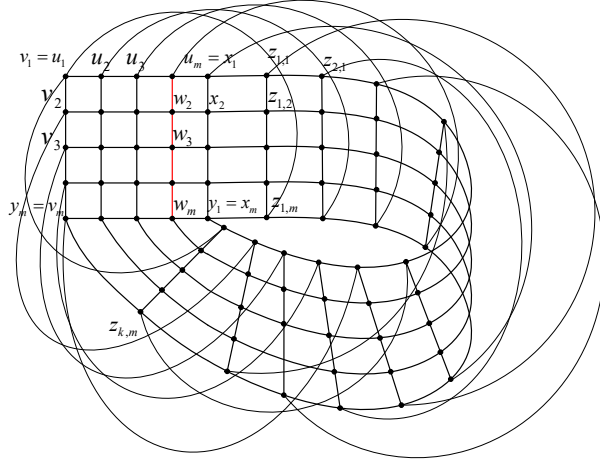


Figure 11: $Q_{k,m,h}$

Theorem 4.3. *The map $Q_{k,m,h}$ with $k \geq 1$ and $m \geq 2$ is l_1 -embeddable if and only if $k = 1$ and $m = 2$.*

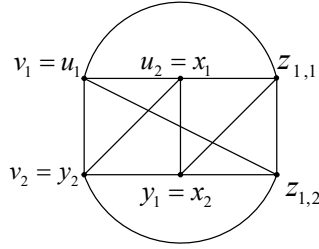


Figure 12: $Q_{1,2,h}$

Proof. By Proposition 3.4, the quadrilateral map on the torus $Q_{3,2,1}$ is l_1 -embeddable. It is easy to check that the graph $Q_{1,2,h}$ in Figure 12 is isomorphic to the map $Q_{3,2,1}$. Hence, the map $Q_{1,2,h}$ is l_1 -embeddable.

Next, we will prove that all maps $Q_{k,m,h}$ except for $\{Q_{1,2,h}\}$ is not l_1 -embeddable.

Case 1. $k \geq 2$ and $m = 2$.

When $k \geq 2$ and $m = 2$, define the mapping $f : V(Q_{k,2,h}) \rightarrow V(Q_{k+2,2,r})$ as

$$\begin{aligned} f(u_1 = v_1) &= a_{1,0}, f(u_2 = x_1) = a_{1,1}, f(y_1 = x_2) = a_{2,1}, \\ f(y_2 = v_2) &= a_{2,0}, f(z_{1,i}) = a_{1,i+1}, f(z_{2,i}) = a_{2,i+1} \quad (1 \leq i \leq k). \end{aligned}$$

It is straightforward to verify that the mapping f is an isomorphism between $Q_{k,2,h}$ and $Q_{k+2,2,2}$. By Theorem 3.1, maps $Q_{k+2,2,2}$ are not l_1 -embeddable. It follows that maps $Q_{k,2,h}$ are not l_1 -embeddable.

Case 2. $k \geq 1$ and $m \geq 3$.

As shown in Figure 10, every face cycle of the map $Q_{k,m,h}$ is isometric, and by Lemma 2.5,

$$l(w_1w_2) = l(x_1x_2) = l(z_{1,1}z_{1,2}) = l(z_{2,1}z_{2,2}) = \cdots = l(z_{k,1}z_{k,2}) = l(w_mx_m) = \cdots = l(w_2x_2).$$

But w_1w_2 and w_2x_2 are adjacent, which contradicts Lemma 2.3. So, the map $Q_{k,m,h}$ with $k \geq 1$ and $m \geq 3$ are not l_1 -embeddable. $\square$

In summary, from Theorems 4.1, 4.2, 4.3, we can obtain the following theorem.

Theorem 4.4. *Among the regular quadrilateral maps on the Klein bottle, only $Q_{2,2,a}$, $Q_{4,2,a}$, $Q_{2,m,b}$, $Q_{1,2,h}$ are l_1 -embeddable.*

5 Conclusion

In this article, we checked whether the regular quadrilateral tilings on the torus and the Klein bottle are l_1 -embeddable. It is found that although there are few classes of tilings are l_1 -embeddable, there are infinitely many l_1 -graphs, which are very different from the hexagonal tilings on the torus and the Klein bottle which are determined by Deza and Shpectorov [7].

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